

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Department of Computer Engineering

Project Progress Report I

Project Title	Integrated Business Operations Platform (IBOP)
Project Members	1. Mann Dutiya 2. 3. 4.
Reporting period	15th January,2026 - 15th April,2026
Section One: Problem Definition	
The project focuses on addressing the challenges faced by small and medium enterprises due to the use of fragmented and disconnected systems for managing business operations such as sales, inventory, finance, and human resources. These systems often lead to data inconsistency, redundant manual work, lack of real-time visibility, and inefficient decision-making. The scope of this project is to design and develop a web-based Integrated Business Operations Platform (IBOP) that centralizes key business functions into a single system. The platform aims to streamline workflows, ensure data consistency, and provide real-time operational insights through an intuitive and modular interface.	
Section Two: Activities and Progress	
System Design: A modular ERP-like architecture was designed, separating the system into distinct components including frontend interface, application logic, and data management. This approach ensures scalability and maintainability. Frontend Development: The user interface was developed using React (Vite) and styled with Tailwind CSS to create a responsive and modular dashboard. The system includes multiple views such as Dashboard, Inventory Management, Sales Module, and Sales Records. Core Module Development: - Inventory Module: Allows addition, updating, and visualization of product data including stock and pricing. - Sales Module: Enables creation of transactions with fields such as product, quantity, rate, and discount. - Automatic Calculations: The system computes total and final amounts dynamically during transactions. The current system operates as a frontend prototype using local state, with the architecture designed to support future backend integration.	
Section Three: Financial Statement	

NA

Section Four: Outputs and Deliverables

The primary deliverable of this project is a functional prototype of the Integrated Business Operations Platform (IBOP), demonstrating the integration of core business modules within a unified system.

The system includes:

- A modular dashboard interface for navigation across different business functions
- Inventory management module for product tracking and stock handling
- Sales module with dynamic transaction creation and validation
- Real-time synchronization between sales and inventory

These deliverables collectively represent a working proof-of-concept for a scalable ERP-like system.

Section Five: Outcomes and Lessons Learned

The project highlighted the significance of integrating multiple business modules into a unified system, where linking sales and inventory ensured data consistency while minimizing manual intervention. Adopting a modular design approach proved effective in separating concerns, making the system easier to scale and extend with additional functionalities such as finance and HR modules. Implementing the solution within a real-world business environment offered valuable insights into practical workflows and usability requirements, emphasizing the need for systems that align with actual operational processes. Additionally, the project reinforced the importance of user experience, particularly the need to design simple, intuitive interfaces that can be easily used by non-technical users in business settings.

Section Six: Dissemination

NA

Section Seven: Risks, Issues and Challenges and Constraints in all context

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Technical Constraints: The current system operates without a backend, limiting persistent data storage and multi-user functionality. This poses a scalability challenge for real-world deployment.

System Complexity: Ensuring accurate synchronization between sales and inventory required careful handling of state and validation logic.

Data Integrity Risks: Without a centralized database, there is a risk of data inconsistency in extended use cases.

Future Risks: Expanding the system to include finance, HR, and multi-user access will increase system complexity and require robust backend architecture.

Section Eight: Institutional & Project Partner Issues

The project is being developed in collaboration with a real business (Shrinath Enterprises). There were no major issues in coordination. The collaboration helped in understanding real-world business workflows and requirements.

Section Nine: Next Steps

Following the successful development of the frontend prototype, the next stages of the project will involve:

- Backend Integration: Implementing a backend using Node.js and integrating a database (MongoDB/PostgreSQL) for persistent storage
- Authentication System: Adding user authentication and role-based access control (Admin/User)
- Module Expansion: Developing additional modules such as Finance and Human Resource Management
- Analytics Dashboard: Implementing data visualization and reporting features for business insights
- System Testing: Conducting functional and performance testing for reliability and scalability

Comments/Suggestions:

Mentors Signature

Expert's Signature